Any 2 × 2 Hermitian matrix **L** can be written as a sum of four terms,

$$\mathbf{L} = a\sigma_x + b\sigma_y + c\sigma_z + dI,$$

where *a*, *b*, *c*, and *d* are real numbers.

Exercise 5.1: Verify this claim.

The unit operator *I* is officially an observable because it is Hermitian, but it's a very boring one. There is only one possible value this trivial observable can have, namely 1,and every state-vector is an eigenvector. If we ignore *I*, then the most general observable is a superposition of the three spin components σ_x,σ_y, and σ_z. Can any pair of spin components be simultaneously measured? Only if they commute. But it is easy to calculate the commutators for these spin components. Just use the matrix representation to multiply them in both orders, and then subtract.

The commutation relations we listed in Eqs. 4.26,

$$\begin{aligned}
[\sigma_x, \sigma_y] &= 2i\sigma_z \\
[\sigma_y, \sigma_z] &= 2i\sigma_x \\
[\sigma_z, \sigma_x] &= 2i\sigma_y,
\end{aligned}$$

tell us straightaway that no two spin components can be simultaneously measured, because the right-hand sides are not zero. In fact, no two components of the spin along any axes can be simultaneously measured.

5.3 THE UNCERTAINTY PRINCIPLE

Uncertainty is one of the hallmarks of quantum mechanics, but it is not always the case that the result of an experiment is uncertain. If a system is in